Submission CALC1-FINAL-SUB02

Question CALC1-FINAL-Q01

work / setup:

1. $f'(x)=2x(3x-1)+(x^2+2)3$

crossed-out arithmetic: [smudged]

FINAL (maybe): 15 plus 1

Question CALC1-FINAL-Q02

work / setup:

1. $\int 3x\,dx = (3/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 37.5 plus 1

Question CALC1-FINAL-Q03

work / setup:

1. $2w+2l=20 \Rightarrow l=10-w$
2. $A(w)=w(10-w)$; at $w=7$, $A=21$

crossed-out arithmetic: [smudged]

FINAL (maybe): $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ plus 1

Question CALC1-FINAL-Q04

work / setup:

1. $x^2-25=(x-5)(x+5)$
2. cancel $x-a$ only after identifying $x\neq a$

crossed-out arithmetic: [smudged]

FINAL (maybe): 10 plus 1

Question CALC1-FINAL-Q05

work / setup:

1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).

crossed-out arithmetic: [smudged]

FINAL (maybe): critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum plus 1

Question CALC1-FINAL-Q06

work / setup:

1. $f'(x)=2x(3x-1)+(x^2+4)3$

crossed-out arithmetic: [smudged]

FINAL (maybe): 11 plus 1

Question CALC1-FINAL-Q07

work / setup:

1. $\int 3x\,dx = (3/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 6 plus 1

Question CALC1-FINAL-Q08
work / setup:
1. $2w+2l=14 \Rightarrow l=7-w$
2. $A(w)=w(7-w)$; at $w=4$, $A=12$
crossed-out arithmetic: [smudged]
FINAL (maybe): $A(w)=w(7-w)$; $A(4)=12$; check $A'(w)=0$ plus 1

Question CALC1-FINAL-Q09
work / setup:
1. $x^2-25=(x-5)(x+5)$
2. cancel $x-a$ only after identifying $x\neq a$
crossed-out arithmetic: [smudged]
FINAL (maybe): 10 plus 1

Question CALC1-FINAL-Q10
work / setup:
1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).
crossed-out arithmetic: [smudged]
FINAL (maybe): critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum plus 1